

NEWARK INTL, NJ, USA

WMO: 725020

Lat: 40.683N

Lon: 74.169W

Elev: 7

StdP: 14.69

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14734

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 12.1	(c) 16.2	(d) -6.2	(e) 3.9	(f) 15.9	(g) -2.0	(h) 4.9	(i) 20.3	(j) 28.4	(k) 29.5	(l) 26.4	(m) 30.1	(n) 13.9	(o) 280	(p) 0.527

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.1	94.0	74.3	91.1	72.8	88.3	71.9	77.5	88.2	76.1	85.4	74.9	83.3	12.4	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.6	129.3	81.9	73.3	123.8	80.4	72.1	118.8	79.7	40.8	88.2	39.4	85.8	38.2	83.0	87.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.1	22.1	19.5	7.1	99.2	5.2	3.3	3.3	101.5	0.3	103.5	-2.6	105.3	-6.4	107.8
			4.9	79.8	4.8	2.1	1.5	81.3	-1.3	82.5	-4.0	83.6	-7.5	85.1
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	55.8	32.6	35.3	42.7	53.7	63.3	72.8	78.1	76.5	69.5	58.0	47.2	38.4
	DBStd	17.56	10.06	9.25	8.99	8.25	7.50	6.61	5.30	5.01	6.33	7.60	8.18	8.73
	HDD50	1810	543	419	260	48	1	0	0	0	16	148	375	
	HDD65	4646	1003	833	691	352	124	13	0	1	27	241	535	826
	CDD50	3926	5	6	35	160	415	683	873	821	586	264	64	14
	CDD65	1285	0	0	1	14	72	246	408	356	163	24	1	0
	CDH74	11297	0	1	17	159	675	2160	4026	3066	1058	132	3	0
	CDH80	4152	0	0	2	45	230	802	1682	1084	283	24	0	0
(14) Wind	WSAvg	9.7	10.7	10.8	11.1	10.5	9.2	9.0	8.7	8.2	8.7	9.3	9.8	10.3
(15) Precipitation	PrecAvg	43.7	3.3	2.9	4.0	3.7	4.1	3.4	4.4	4.0	3.9	2.9	3.6	3.5
	PrecMax	65.5	10.1	5.0	11.1	11.1	10.2	10.5	10.0	10.6	9.4	6.8	11.5	9.4
	PrecMin	26.1	0.4	0.5	1.1	0.9	0.5	0.4	0.9	0.5	1.0	0.2	0.5	0.6
	PrecStd	8.4	2.0	1.1	2.0	2.0	2.1	1.9	2.5	2.2	2.2	1.5	2.3	1.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.3	65.2	75.6	86.2	91.3	96.0	98.9	96.2	91.6	83.2	71.9	66.8
		MCWB	57.2	54.8	60.8	65.0	70.0	74.8	75.4	75.7	71.6	67.5	61.8	60.0
	2%	DB	56.9	58.1	66.9	78.5	86.3	91.5	94.9	92.1	86.6	77.6	67.3	60.0
		MCWB	52.9	50.1	54.4	60.7	68.4	73.2	74.4	74.0	70.9	65.9	59.3	54.8
	5%	DB	50.8	52.7	60.8	72.6	81.7	88.1	91.7	88.9	83.0	73.6	63.5	55.2
		MCWB	46.0	45.1	50.4	57.8	65.7	71.4	73.3	72.8	69.4	64.1	56.5	50.6
	10%	DB	46.4	48.1	56.2	67.4	76.8	84.6	88.5	85.8	79.9	70.1	59.8	51.0
		MCWB	41.8	41.8	47.6	54.3	63.3	69.0	72.6	71.1	68.2	61.8	53.9	45.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.3	59.1	62.2	67.4	73.2	77.7	79.8	78.9	76.4	72.0	65.4	61.2
		MCDB	61.8	63.6	72.1	80.3	85.7	90.5	93.0	90.8	84.7	76.8	69.2	64.9
	2%	WB	53.5	51.6	57.2	63.6	70.7	75.4	77.9	77.1	74.8	69.2	61.7	56.0
		MCDB	55.8	56.8	63.5	74.6	82.4	86.4	89.1	86.9	81.0	74.8	65.7	59.3
	5%	WB	46.6	46.2	52.3	60.3	68.6	73.6	76.4	75.7	73.1	66.4	58.2	51.8
		MCDB	49.8	50.9	58.8	69.7	78.2	83.9	86.2	83.6	79.0	71.8	61.8	54.8
	10%	WB	42.3	42.4	48.3	56.7	65.7	72.0	75.0	74.4	71.1	63.5	54.9	46.6
		MCDB	45.9	47.3	54.9	64.4	74.3	81.3	84.0	81.8	77.2	68.6	58.8	49.8
(35) Mean Daily Temperature Range	5% DB	MDBR	13.2	14.4	15.6	17.1	17.2	16.4	16.1	15.3	15.5	15.6	14.5	12.5
		MCDBR	18.7	20.7	22.4	24.7	23.8	20.8	19.9	18.8	19.1	19.5	18.5	17.6
		MCWBR	16.2	15.6	14.4	12.6	10.9	8.6	7.1	7.4	8.5	10.6	13.3	15.5
	5% WB	MCDBR	17.9	19.1	20.8	22.0	21.2	18.4	16.9	15.7	14.5	16.1	15.8	16.5
		MCWBR	16.9	16.3	15.8	13.4	11.2	8.9	7.2	7.3	8.4	11.3	14.1	16.3
(40) Clear-Sky Solar Irradiance	taub		0.315	0.328	0.362	0.385	0.432	0.493	0.494	0.481	0.403	0.364	0.332	0.314
	taud		2.459	2.422	2.389	2.357	2.256	2.124	2.135	2.163	2.352	2.473	2.518	2.521
	Ebn at Noon		268	281	282	281	269	252	251	250	264	265	260	259
	Edh at Noon		25	30	34	38	43	49	48	45	35	28	23	22
(44) All-Sky Solar Radiation	RadAvg		592	879	1180	1518	1683	1848	1887	1637	1365	951	668	505
	RadStd		50	89	98	148	177	168	92	106	136	96	59	49

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (61 neighbors)		+0.72	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+211	+122

Nomenclature: See separate page